

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 8: Synthesis and explanation

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	B	A	C	C	C	A	B	D	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share prediction on classifying 6 unfamiliar organisms from Kericho, Kisumu or Mombasa

Q1. Learners from a school in Kisumu receive 6 unfamiliar organisms and must group them exactly as a school in Kericho does. What would help them most to get the same groups?

- A. Use the same observable characteristics and a shared dichotomous key**
- B. Group them by size, from the biggest to the smallest
- C. Group them by how useful each one is as food
- D. Group them by colour so that similar-looking ones are together

Correct answer: A. Shared, observable characteristics and a key let any learner reach the same groups. Size, colour and use as food vary from person to person and give different groupings, as seen in Lesson 1.

USE AFTER: OBSERVE PHASE — Final classification challenge: one-line note on viruses (why they are not placed in either kingdom)

Q2. Why are viruses not placed in Kingdom Plantae or Kingdom Animalia?

- A. Viruses are too small to be seen with any instrument
- B. Viruses are not made of cells, so they do not fit into either kingdom**
- C. Viruses only live in water and not on land
- D. Viruses are single cells that have no nucleus

Correct answer: B. Viruses are non-cellular. Plants and animals are made of cells, so viruses cannot be placed in either kingdom. The idea that 'viruses are cells' is a common misconception.

USE AFTER: OBSERVE PHASE — Final classification challenge, Task 1: sorting cards into Divisions (Bryophyta, Pteridophyta, Spermatophyta)

Q3. A fern is in Division Pteridophyta and moss is in Division Bryophyta. Which feature separates the fern from the moss?

- A. The fern has vascular tissue, but the moss does not**
- B. The moss makes seeds, but the fern makes spores
- C. The fern has flowers, but the moss does not
- D. The moss has vascular tissue, but the fern does not

Correct answer: A. Ferns have vascular tissue and reproduce by spores. Mosses lack vascular tissue. Neither makes seeds or flowers.

USE AFTER: OBSERVE PHASE — Final classification challenge, Task 1: sorting maize and bean into monocot and dicot

Q4. A plant has seeds enclosed in a fruit, leaves with net veins, and seeds with two cotyledons. Which group does it belong to?

- A. Gymnospermae
- B. Bryophyta
- C. Angiospermae, dicotyledon**
- D. Angiospermae, monocotyledon

Correct answer: C. Seeds enclosed in a fruit mean Angiospermae. Net veins and two cotyledons mean a dicotyledon, like the bean. Maize has parallel veins and one cotyledon, so it is a monocotyledon.

USE AFTER: OBSERVE PHASE — Final classification challenge, Task 1: sorting into Phyla and Classes (Chordata and Arthropoda)

Q5. A mosquito has a jointed exoskeleton, three pairs of jointed legs and no backbone. Which phylum does it belong to, and why?

- A. Arthropoda, because it is small and harmful to people
- B. Chordata, because it is an animal that can fly
- C. Arthropoda, because it has a jointed exoskeleton and jointed legs**
- D. Chordata, because it has three pairs of legs

Correct answer: C. Arthropods have a jointed exoskeleton and jointed legs. Chordates have a backbone, which the mosquito lacks. Size and harmfulness are not classification characteristics.

USE AFTER: OBSERVE PHASE — Final classification challenge, Task 2: build or complete a dichotomous key for at least 6 organisms and exchange it with another group

Q6. Which pair of statements is best for one step in a dichotomous key for insects?

- A. Is liked by learners / is not liked by learners
- B. Is big / is small
- C. Has wings / has no wings**
- D. Looks beautiful / does not look beautiful

Correct answer: C. A key needs paired statements that are observable and clear, so any user gets the same answer. 'Has wings / no wings' can be checked by anyone. Beauty, size and liking are opinions or vary between people.

USE AFTER: EXPLAIN PHASE — Final Explanation: Claim-Evidence-Reasoning (CER) answering 'How can we identify and classify living organisms accurately?'

Q7. A learner writes: 'Maize has one cotyledon and parallel veins, but bean has two cotyledons and net veins.' Which reasoning best explains why this evidence helps to classify them accurately?

- A. Any learner can observe and check these features, so everyone gets the same groups**
- B. Maize is eaten more than bean, so it needs its own group
- C. The teacher said that maize and bean are different, so they are
- D. Maize is bigger than bean, so they must be separate groups

Correct answer: A. Good reasoning links evidence to the principle that shared, observable characteristics give the same grouping for any learner. Size, use as food and authority do not.

USE AFTER: DQB PHASE — Driving Question Board: writing at least one new question beyond the classroom

Q8. After sorting the Driving Question Board, a group wants to add a new question that goes beyond the classroom. Which question is the best fit?

- A. Which organism in our collection is the prettiest?
- B. How can a dichotomous key help identify crop pests in Nakuru?**
- C. Why do we have a Driving Question Board?
- D. Is a goat bigger than a chicken?

Correct answer: B. The best new question applies classification knowledge to a real situation such as farming and can be investigated. The others are opinions, simple comparisons or about classroom routine.

USE AFTER: MODEL BUILDING PHASE — Final model: 'Can your model place a new organism you have never seen, say a tilapia or a cypress tree?'

Q9. A learner finds a tilapia from Lake Victoria. It has a backbone, gills and fins. Where should the learner's final classification model place it?

- A. Kingdom Animalia, Phylum Arthropoda
- B. Kingdom Animalia, Phylum Chordata, Class Birds
- C. Kingdom Plantae, Division Spermatophyta
- D. Kingdom Animalia, Phylum Chordata, Class Fish**

Correct answer: D. Animals with a backbone are in Phylum Chordata, and a fish with gills and fins belongs to the fish class. Arthropods lack a backbone, and birds are a different class of Chordata.

Q10. Which statement about the importance of viruses to Kenyans is correct?

- A. Viruses can cause diseases such as measles and maize streak, but are also used in vaccines and biotechnology**
- B. Viruses affect only people and never crops
- C. Viruses make food for plants by photosynthesis
- D. Viruses only cause disease and have no use

Correct answer: A. Viruses cause harm in health (for example measles) and agriculture (maize streak virus), but are also used in vaccines and biotechnology.